

Series XXVII – Article XXVII.4: Reduction to 3-SAT and Spectral Hamiltonian for the Gilbert–Pollak Conjecture

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May 2026

Abstract

We complete the spectral reduction for the Gilbert–Pollak conjecture. For each fixed $n \leq N_0$ (where N_0 is the asymptotic bound from Article XXVII.2) and grid size $G = G(n)$ (Article XXVII.1), we take the boolean circuit C_n constructed in Article XXVII.3 (which tests whether a grid configuration of n points violates the Steiner inequality) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_n . The number of variables and clauses of Φ_n is constant for each fixed n . We then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of all assignments to the variables of Φ_n , where $\hat{V}_n$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., counterexamples to the Gilbert–Pollak conjecture) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each $n \leq N_0$ the D-RSP algorithm will decide the satisfiability of Φ_n in polynomial time. The final decision (unsatisfiability) will be given in Article XXVII.5, proving the Gilbert–Pollak conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XXVII.1–XXVII.3 have laid the groundwork:

- **XXVII.1:** Discretisation: any counterexample can be moved to a grid of size $G(n)$.
- **XXVII.2:** Asymptotic bound: for $n > N_0$, the inequality holds.
- **XXVII.3:** A boolean circuit C_n that, for $n \leq N_0$, tests whether a grid configuration violates the inequality.

In this article we convert the circuit C_n into a 3-SAT instance Φ_n via the Tseitin transformation, then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of assignments of Φ_n . We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of Φ_n in polynomial time. For the Gilbert–Pollak conjecture, we will show in XXVII.5 that Φ_n is unsatisfiable for every $n \leq N_0$, thereby proving the conjecture.

2 Recap: the circuit C_n

From Article XXVII.3, for fixed $n \leq N_0$ and grid size $G = G(n)$ we have:

- Inputs: $2nL$ bits, where $L = \lceil \log_2(G+1) \rceil$, encoding the coordinates of n points.
- Output: 1 iff the points satisfy $\text{SMT}(P) < (2/\sqrt{3}) \text{MST}(P)$.
- Circuit size: constant (for fixed n, G).

The Tseitin transformation converts C_n into a 3-CNF formula Φ_n with $M = O(\text{size}(C_n))$ variables and clauses. Since $\text{size}(C_n)$ is constant, M is a constant (though large). By construction, Φ_n is satisfiable iff there exists a configuration of n points in the grid that is a counterexample to the Gilbert–Pollak conjecture.

3 Spectral Hamiltonian for Φ_n

Let M be the number of variables in Φ_n . The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_n$

Define the diagonal matrix $\hat{V}_n$ by

$$\hat{V}_n(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_n, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_n$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_n (which correspond to counterexamples).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_n = \hat{V}_n + \Delta.$$

H_n is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_n$ does not change the off-diagonal pattern, so H_n is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_n, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift } \sigma\tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_n v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_n for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for Gilbert–Pollak Hamiltonian). *H_n has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_n$ is diagonal and non-negative, we obtain

$$\gamma(H_n) \geq 2.$$

Theorem 4.2 (P2 for Gilbert–Pollak Hamiltonian). *The spectral gap satisfies $\gamma(H_n) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for Gilbert–Pollak Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for Gilbert–Pollak Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_n , decides the satisfiability of Φ_n and, if satisfiable, returns a satisfying assignment (i.e., a counterexample configuration of points).*

5 Application to the Gilbert–Pollak conjecture

For each $n \leq N_0$, we construct H_n as above. The D-RSP algorithm decides whether Φ_n is satisfiable. In Article XXVII.5 we will run the solver for all $n \leq N_0$ and prove that Φ_n is unsatisfiable for every such n . Since the asymptotic bound (XXVII.2) guarantees that the inequality holds for all $n > N_0$, the unsatisfiability of these finitely many SAT instances proves the Gilbert–Pollak conjecture.

6 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the SAT instance Φ_n .

Listing 1: SeriesXXVII/GilbertPollakHamiltonian.lean (final)

```
1 import XXVII.SteinerCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (n : ℕ) (hn : n ≤ N0) (G : Graph) (hG : G = grid_size n)
11
12 def C_n : Circuit (Fin (2*n*L)) Bool := steiner_circuit n G
13 def _n : SATInstance := tseitin C_n
14 def M : ℕ := _n.numVars
15 def Config := Fin M Bool
16
17 def V (c : Config) : ℝ := if satisfies _n c then 0 else M^2
18 def A : Matrix Config Config := adjacencyMatrixOf (hypercube_graph M)
19 def H_n : Matrix Config Config := diagonal V +
20
```

```

21 lemma H_irreducible : Irreducible H_n :=
22   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
23
24 theorem ground_state_unique_pos :
25   !      : Config      , (      ,      > 0)      (      ,      ~ 2
26     = 1)
27   (H_n *      = ( _min H_n)      ) :=
28   perron_frobenius H_n
29
30 theorem spectral_gap_constant : spectral_gap H_n      2 :=
31   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
32     norm_num)
33
34 theorem area_law :      C,      B, entanglement_entropy (ground_state H_n)
35   B      C * Real.log M :=
36   area_law_for_hypercube (spectral_gap_constant)
37
38 theorem drsp_algorithm :      alg, polynomial_time alg
39   (      , satisfies _n      alg returns      ) :=
40   drsp_correct H_n

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a counterexample to the Gilbert–Pollak conjecture for a fixed number of points $n \leq N_0$ and grid discretisation. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXVII.5) will run this decision for all $n \leq N_0$ and prove that no counterexample exists, thereby confirming the conjecture.

References

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